Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

Summary of changes:

1. 2019enh3: CF is calculated based on the default value of TOXE, but not using updated value of TOXE (**ADI, Infineon**)
2. 2021enh1: Description of U0 protection in BSIM4 manual (**Primarius**)
3. 2021enh2: Checks on BSIM4phi, BSIM4phieot, and BSIM4tempeot (**ADI**)
4. 2021enh3: Typo in intrinsic carrier concentration equation in the technical manual (**ADI**)
5. 2022enh1: Typo in the BSIM4 technical manual for mobility model – MOBMOD = 3 (**ADI**)
6. 2022enh2: ids0mult parameter (MULT_I, MULT_Q, MULT_FN) (**GF/Cadence**)
7. 2022bug1: Correction of if condition for BSIM4PhiBSWS parameter (**ADI**)
8. 2022bug3: Missing new parameters (GIDLCLAMP / IDovVDSC) in b4mpar.c (**UCB/NGSPICE**)
9. 2022bug4: Complete documentation of tempMod = 2 and 3 (**ADI**)
10. 2023enh1: Inclusion of comprehensive QA test for MULT check (**Cadence/BSIM group**)
11. 2023bug1: BSIM4 non-zero terminal current with all terminal voltage zero (**NXP/Cadence**)
12. 2023bug2: Negative Cgg/Cbg at high Vdd, high back bias and negative keta (**TI/Cadence**)
13. 2024enh1: Missing Details of Length/Width/Cross dependencies of binnable model parameters in BSIM4_4.8.3_beta1 Technical Manual (**GF**)
14. 2024bug1: Making bug-fix (2023bug2) backward compatible (**ADI/Cadence**)
15. 2024bug2: Interpretation for default value of dW & dL parameters in BSIM4_4.8.3_beta1 Technical Manual (**GF**)
16. 2024bug3: Missing lines in b4mask.c and b4ask.c to handle a few parameters (**ADI**).
17. 2024bug4: BSIM4 version check (**ADI**).
18. 2024bug5: Code update for developer's names (**ADI**).

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1) 2019enh3: CF is calculated based on the default value of TOXE, but not using updated value of TOXE (ADI)

Solution: Calculation of CF in b4temp.c file is moved out from if conditions on BSIM4mtrlMod and is now calculated based on the updated value of TOXE.

Old Code

b4temp.c

```
if(model->BSIM4mtrlMod == 0)
{
    if ((model->BSIM4toxeGiven) && (model->BSIM4toxpGiven) && (model->BSIM4dtoxGiven)
    && (model->BSIM4toxe != (model->BSIM4toxp + model->BSIM4dtox)))
    {   printf("Warning: toxe, toxp and dtox all given and toxe != toxp + dtox; dtox ignored.\n");
    }
    else if ((model->BSIM4toxeGiven) && (!model->BSIM4toxpGiven))
    { model->BSIM4toxp = model->BSIM4toxe - model->BSIM4dtox;
    }
    else if ((!model->BSIM4toxeGiven) && (model->BSIM4toxpGiven))
    {
        model->BSIM4toxe = model->BSIM4toxp + model->BSIM4dtox;
        if (!model->BSIM4toxmGiven)             /* v4.7 */
            model->BSIM4toxm = model->BSIM4toxe;
    }
      if (!model->BSIM4cfGiven)             /* v4.8.2 */
           model->BSIM4cf = 2.0 * model->BSIM4epsrox * EPS0 / PI
            * log(1.0 + 0.4e-6 / model->BSIM4toxe);
}

else if(model->BSIM4mtrlCompatMod != 0) /* v4.7 */
{
    T0 = model->BSIM4epsrox / 3.9;
    if ((model->BSIM4eotGiven) && (model->BSIM4toxpGiven) && (model->BSIM4dtoxGiven)
    && (ABS(model->BSIM4eot * T0 - (model->BSIM4toxp + model->BSIM4dtox)) > 1.0e-20))
    {
        printf("Warning: eot, toxp and dtox all given and eot * EPSROX / 3.9 != toxp + dtox; dtox ignored.\n");
    }
    else if ((model->BSIM4eotGiven) && (!model->BSIM4toxpGiven))
      model->BSIM4toxp = T0 * model->BSIM4eot - model->BSIM4dtox;
    else if ((!model->BSIM4eotGiven) && (model->BSIM4toxpGiven)){
      model->BSIM4eot = (model->BSIM4toxp + model->BSIM4dtox) / T0;
        if (!model->BSIM4toxmGiven)
            model->BSIM4toxm = model->BSIM4eot;
    }
}

if(model->BSIM4mtrlMod)
  {
    epsrox = 3.9;
    toxe = model->BSIM4eot;
    epssub = EPS0 * model->BSIM4epsrsub;
  }
else
  {
    epsrox = model->BSIM4epsrox;
    toxe = model->BSIM4toxe;
    epssub = EPSSI;
  }
```

New Code

b4temp.c

```
if(model->BSIM4mtrlMod == 0)
{
    if ((model->BSIM4toxeGiven) && (model->BSIM4toxpGiven) && (model->BSIM4dtoxGiven)
    && (model->BSIM4toxe != (model->BSIM4toxp + model->BSIM4dtox)))
    {   printf("Warning: toxe, toxp and dtox all given and toxe != toxp + dtox; dtox ignored.\n");
    }
    else if ((model->BSIM4toxeGiven) && (!model->BSIM4toxpGiven))
    { model->BSIM4toxp = model->BSIM4toxe - model->BSIM4dtox;
    }
    else if ((!model->BSIM4toxeGiven) && (model->BSIM4toxpGiven))
    {
        model->BSIM4toxe = model->BSIM4toxp + model->BSIM4dtox;
        if (!model->BSIM4toxmGiven)             /* v4.7 */
            model->BSIM4toxm = model->BSIM4toxe;
    }
}

else if(model->BSIM4mtrlCompatMod != 0) /* v4.7 */
{
    T0 = model->BSIM4epsrox / 3.9;
    if ((model->BSIM4eotGiven) && (model->BSIM4toxpGiven) && (model->BSIM4dtoxGiven)
    && (ABS(model->BSIM4eot * T0 - (model->BSIM4toxp + model->BSIM4dtox)) > 1.0e-20))
    {
        printf("Warning: eot, toxp and dtox all given and eot * EPSROX / 3.9 != toxp + dtox; dtox ignored.\n");
    }
    else if ((model->BSIM4eotGiven) && (!model->BSIM4toxpGiven))
      model->BSIM4toxp = T0 * model->BSIM4eot - model->BSIM4dtox;
    else if ((!model->BSIM4eotGiven) && (model->BSIM4toxpGiven)){
      model->BSIM4eot = (model->BSIM4toxp + model->BSIM4dtox) / T0;
        if (!model->BSIM4toxmGiven)
            model->BSIM4toxm = model->BSIM4eot;
    }
}

if(model->BSIM4mtrlMod)
  {
    epsrox = 3.9;
    toxe = model->BSIM4eot;
    epssub = EPS0 * model->BSIM4epsrsub;
  }
else
  {
    epsrox = model->BSIM4epsrox;
    toxe = model->BSIM4toxe;
    epssub = EPSSI;
  }
if (!model->BSIM4cfGiven)
     model->BSIM4cf = 2.0 * epsrox * EPS0 / PI
      * log(1.0 + 0.4e-6 / toxe);
```

2) 2021enh1: Description of U0 protection in BSIM4 manual (Primarius)

Solution: Following description about unit conversion of U0 is added to note section of the appendix in the technical manual.

“If user specifies U0 > 1 then, an implicit unit conversion takes place and the user provided U0 value gets multiplied by 1e-4.”

3) 2021enh2: Checks on BSIM4phi, phieot, BSIM4tempeot (ADI)

Issue: The square root of BSIM4phi will give a math error even before running the checks on BSIM4phi, defined in *b4check.c* file.

Solution: Check on Phi is moved from *b4check.c* to *b4temp.c* before BSIM4sqrtPhi calculation.

New Code

b4temp.c

```
pParam->BSIM4phi = Vtm0 * log(pParam->BSIM4ndep / ni)
         + pParam->BSIM4phin + 0.4;

if (pParam->BSIM4phi <= 0.0)
{   fprintf(fplog, "Fatal: Phi = %g is not positive. Please check Phin and Ndep\n",
            pParam->BSIM4phi);
    fprintf(fplog, "       Phin = %g  Ndep = %g \n",
            pParam->BSIM4phin, pParam->BSIM4ndep);
    printf("Fatal: Phi = %g is not positive. Please check Phin and Ndep\n",
            pParam->BSIM4phi);
    printf("       Phin = %g  Ndep = %g \n",
            pParam->BSIM4phin, pParam->BSIM4ndep);
    Fatal_Flag = 1;
}

    pParam->BSIM4sqrtPhi = sqrt(pParam->BSIM4phi);
```

Issue: There is no check on phieot (defined for MTRLMOD=1); the square root of phieot will give a math error.

Solution: Check on phieot is added in *b4temp.c* file.

New Code

b4temp.c

```
 phieot = Vtm0eot * log(pParam->BSIM4ndep / ni)
        + pParam->BSIM4phin + 0.4;

if (phieot <= 0.0)
{   fprintf(fplog, "Fatal: phieot = %g is not positive. Please Check Phin and Ndep\n",
            pParam->BSIM4phi);
    fprintf(fplog, " Phin = %g Ndep = %g \n",
            pParam->BSIM4phin, pParam->BSIM4ndep);
    printf("Fatal: phieot = %g is not positive. Please Check Phin and Ndep\n",
            pParam->BSIM4phi);
    printf(" phieot = %g Ndep = %g \n",
            pParam->BSIM4phin, pParam->BSIM4ndep);
    Fatal_Flag = 1;
}
```

Issue: There is no check on BSIM4tempeot (defined for MTRLMOD=1).

Solution: Check on BSIM4tempeot is added in *b4check.c* file.

New Code

b4check.c

```
if(model->BSIM4tempeot <= 0.0)
{    fprintf(fplog, "Fatal: TEMPEOT = %g is not positive.\n",
             model->BSIM4tempeot);
     printf("Fatal: TEMPEOT = %g is not positive.\n", model->BSIM4tempeot);
     Fatal_Flag = 1;
}
```

4) 2021enh3: Typo in intrinsic carrier concentration equation in the technical manual (ADI)

Solution: The intrinsic carrier concentration equation is corrected in the manual.

Old Equation

$$n_i = 1.45e10 \cdot \frac{TNOM}{300.15} \cdot \sqrt{\frac{TNOM}{300.15}} \cdot exp\left[21.5565981 - \frac{qE_g(TNOM)}{2 \cdot k_B \cdot T}\right]$$

New Equation

$$n_i = 1.45e10 \cdot \frac{TNOM}{300.15} \cdot \sqrt{\frac{TNOM}{300.15}} \cdot exp\left[21.5565981 - \frac{qE_g(TNOM)}{2 \cdot k_B \cdot TNOM}\right]$$

5) 2022bug1: Correction of if condition for BSIM4PhiBSWS parameter (ADI)

Solution: The inequality "<=" for parameters BSIM4PhiBSWS, BSIM4PhiBSWD, BSIM4PhiBSWGS and BSIM4PhiBSWGD is replaced by "<".

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Old Code

b4temp.c

```
    model->BSIM4PhiBSWS = model->BSIM4SsidewallJctPotential
                        - model->BSIM4tpbsw * delTemp;
    if (model->BSIM4PhiBSWS <= 0.01)
{   model->BSIM4PhiBSWS = 0.01;
    fprintf(stderr, "Temperature effect has caused pbsws to be less than 0.01. Pbsws is clamped to 0.01.\n");
}
    model->BSIM4PhiBSWD = model->BSIM4DsidewallJctPotential
                        - model->BSIM4tpbsw * delTemp;
    if (model->BSIM4PhiBSWD <= 0.01)
    {   model->BSIM4PhiBSWD = 0.01;
        fprintf(stderr, "Temperature effect has caused pbswd to be less than 0.01. Pbswd is clamped to 0.01.\n");
    }

model->BSIM4PhiBSWGS = model->BSIM4SGatesidewallJctPotential
                        - model->BSIM4tpbswg * delTemp;
    if (model->BSIM4PhiBSWGS <= 0.01)
{   model->BSIM4PhiBSWGS = 0.01;
    fprintf(stderr, "Temperature effect has caused pbswgs to be less than 0.01. Pbswgs is clamped to 0.01.\n");
}
    model->BSIM4PhiBSWGD = model->BSIM4DGatesidewallJctPotential
                        - model->BSIM4tpbswg * delTemp;
    if (model->BSIM4PhiBSWGD <= 0.01)
    {   model->BSIM4PhiBSWGD = 0.01;
        fprintf(stderr, "Temperature effect has caused pbswgd to be less than 0.01. Pbswgd is clamped to 0.01.\n");
    } /* End of junction capacitance */
```

New Code

b4temp.c

```
     model->BSIM4PhiBSWS = model->BSIM4SsidewallJctPotential
                         - model->BSIM4tpbsw * delTemp;
     if (model->BSIM4PhiBSWS < 0.01)
 {   model->BSIM4PhiBSWS = 0.01;
     fprintf(stderr, "Temperature effect has caused pbsws to be less than 0.01. Pbsws is clamped to 0.01.\n");
 }
     model->BSIM4PhiBSWD = model->BSIM4DsidewallJctPotential
                         - model->BSIM4tpbsw * delTemp;
     if (model->BSIM4PhiBSWD < 0.01)
     {   model->BSIM4PhiBSWD = 0.01;
         fprintf(stderr, "Temperature effect has caused pbswd to be less than 0.01. Pbswd is clamped to 0.01.\n");
     }

 model->BSIM4PhiBSWGS = model->BSIM4SGatesidewallJctPotential
                         - model->BSIM4tpbswg * delTemp;
     if (model->BSIM4PhiBSWGS < 0.01)
 {   model->BSIM4PhiBSWGS = 0.01;
     fprintf(stderr, "Temperature effect has caused pbswgs to be less than 0.01. Pbswgs is clamped to 0.01.\n");
 }
     model->BSIM4PhiBSWGD = model->BSIM4DGatesidewallJctPotential
                         - model->BSIM4tpbswg * delTemp;
     if (model->BSIM4PhiBSWGD < 0.01)
     {   model->BSIM4PhiBSWGD = 0.01;
         fprintf(stderr, "Temperature effect has caused pbswgd to be less than 0.01. Pbswgd is clamped to 0.01.\n");
     } /* End of junction capacitance */
```

6) 2022bug3: Missing new parameters (GIDLCLAMP / IDovVDSC) in b4mpar.c (UCB/NGSPICE)

Solution: GIDLCLAMP and IDOVVDSC are introduced in b4mpar.c file.

New Code

b4temp.c

```
case  BSIM4_MOD_GIDLCLAMP :  /* gidlclamp introduced in b4mpar.c */
          mod->BSIM4gidlclamp = value->rValue;
          mod->BSIM4gidlclampGiven = TRUE;
          break;
case  BSIM4_MOD_IDOVVDSC :   /* idovvdsc introduced in b4mpar.c */
          mod->BSIM4idovvdsc = value->rValue;
          mod->BSIM4idovvdscGiven = TRUE;
          break;
```

7) 2022bug4: Incomplete documentation of tempMod = 0,1 and 2,3 (ADI)

Solution: Technical manual is updated for tempMOD 0, 1, 2, and 3.

The following description is added to the note section of technical manual to highlight the reason for difference in the equations for tempMod 0, 1 versus tempMod 2, 3.

"*The equation for tempMod = 0,1 is more physical and complete as it includes length and bias dependence. While tempMod = 2,3 provides backward compatibility*."

8) 2022enh1: Typo in the BSIM4 technical manual for mobility model – MOBMOD = 3 (ADI)

Typo in μ_{eff} equation in the technical manual is corrected in the BSIM4 4.8.3 technical manual.

Old Equation

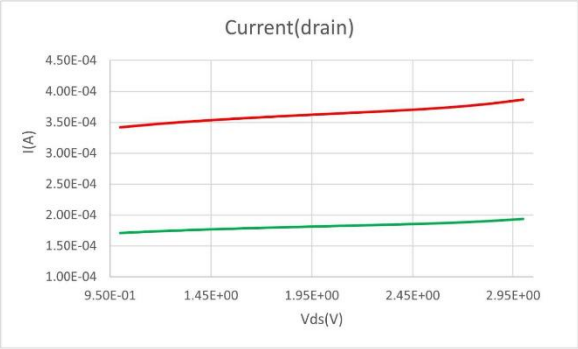
$$\mu_{eff} = \frac{U0.f(L_{eff})}{1+(UA+UC.V_{bseff})\left[\frac{V_{gsteff}+C_0.(VTH0-V_{fb}-\Phi_S)}{6.TOXE}\right]^{EU}+\frac{UD}{0.5\left[1+V_{gsteff}/V_{gsteff,Vth}\right]^{UCS}}}$$

New Equation

$$\mu_{eff} = \frac{U0.f(L_{eff})}{1+(UA+UC.V_{bseff})\left[\frac{V_{gsteff}+C_0.(VTH0-V_{fb}-\Phi_S)}{6.TOXE}*10^{-8}\right]^{EU}+\frac{UD}{\left[0.5\left(1+V_{gsteff}/V_{gsteff,Vth}\right)\right]^{UCS}}}$$

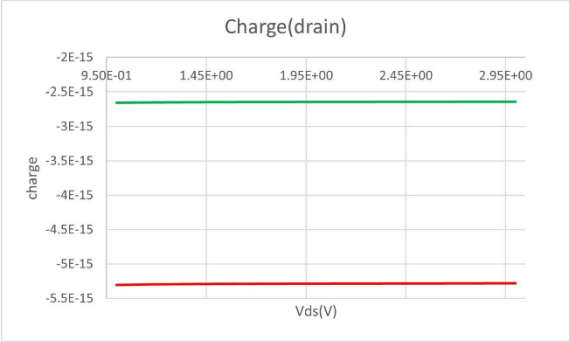
9) 2022enh2: ids0mult parameter (MULT_I, MULT_Q, MULT_FN) (GF/Cadence)

Solution: The mult parameters are introduced to scale all currents, charges, and noise.



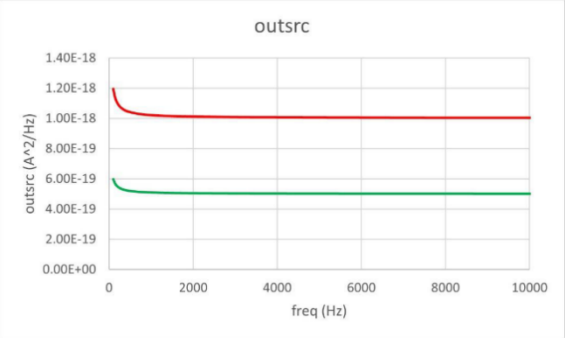
red line: mult_i=2

green line: mult_i=1



red line: mult_q=2

green line: mult_q=1



red line: mult_fn=2

green line: mult_fn=1

They were introduced in the code as follows.

Old Code

b4ask.c

```
case BSIM4_SOURCECONDUCT:
    value->rValue = here->BSIM4sourceConductance;
    return(OK);
case BSIM4_DRAINCONDUCT:
    value->rValue = here->BSIM4drainConductance;
    return(OK);

case BSIM4_CD:
    value->rValue = here->BSIM4cd;
    return(OK);
case BSIM4_CBS:
    value->rValue = here->BSIM4cbs;
    return(OK);
case BSIM4_CBD:
    value->rValue = here->BSIM4cbd;
    return(OK);
case BSIM4_CSUB:
    value->rValue = here->BSIM4csub;
    return(OK);
case BSIM4_IGIDL:
    value->rValue = here->BSIM4Igidl;
    return(OK);

case BSIM4_IGISL:
    value->rValue = here->BSIM4Igisl;
    return(OK);
case BSIM4_IGS:
    value->rValue = here->BSIM4Igs;
    return(OK);
case BSIM4_IGD:
    value->rValue = here->BSIM4Igd;
    return(OK);
case BSIM4_IGB:
    value->rValue = here->BSIM4Igb;
    return(OK);
case BSIM4_IGCS:
    value->rValue = here->BSIM4Igcs;
    return(OK);
case BSIM4_IGCD:
    value->rValue = here->BSIM4Igcd;
    return(OK);

case BSIM4_GM:
    value->rValue = here->BSIM4gm;
    return(OK);
case BSIM4_GDS:
    value->rValue = here->BSIM4gds;
    return(OK);
case BSIM4_GMBS:
    value->rValue = here->BSIM4gmbs;
    return(OK);
case BSIM4_GBD:
    value->rValue = here->BSIM4gbd;
    return(OK);
case BSIM4_GBS:
    value->rValue = here->BSIM4gbs;
    return(OK);
```

```
case BSIM4_QB:
    value->rValue = here->BSIM4qbulk;
    return(OK);
case BSIM4_QG:
    value->rValue = here->BSIM4qgate;
    return(OK);
case BSIM4_QS:
    value->rValue = here->BSIM4qsrc;
    return(OK);
case BSIM4_QD:
    value->rValue = here->BSIM4qdrn;
    return(OK);

case BSIM4_GCRG:
    value->rValue = here->BSIM4gcrg;
    return(OK);

case BSIM4_CGGB:
    value->rValue = here->BSIM4cggb;
    return(OK);
case BSIM4_CGDB:
    value->rValue = here->BSIM4cgdb;
    return(OK);
case BSIM4_CGSB:
    value->rValue = here->BSIM4cgsb;
    return(OK);
case BSIM4_CDGB:
    value->rValue = here->BSIM4cdgb;
    return(OK);
case BSIM4_CDDB:
    value->rValue = here->BSIM4cddb;
    return(OK);
case BSIM4_CDSB:
    value->rValue = here->BSIM4cdsb;
    return(OK);

case BSIM4_CBGB:
    value->rValue = here->BSIM4cbgb;
    return(OK);
case BSIM4_CBDB:
    value->rValue = here->BSIM4cbdb;
    return(OK);
case BSIM4_CBSB:
    value->rValue = here->BSIM4cbsb;
    return(OK);
case BSIM4_CSGB:
    value->rValue = here->BSIM4csgb;
    return(OK);
case BSIM4_CSDB:
    value->rValue = here->BSIM4csdb;
    return(OK);
case BSIM4_CSSB:
    value->rValue = here->BSIM4cssb;
    return(OK);
```

```
case BSIM4_CGBB:
    value->rValue = here->BSIM4cgbb;
    return(OK);
case BSIM4_CDBB:
    value->rValue = here->BSIM4cdbb;
    return(OK);
case BSIM4_CSBB:
    value->rValue = here->BSIM4csbb;
    return(OK);
case BSIM4_CBBB:
    value->rValue = here->BSIM4cbbb;
    return(OK);
case BSIM4_CAPBD:
    value->rValue = here->BSIM4capbd;
    return(OK);
case BSIM4_CAPBS:
    value->rValue = here->BSIM4capbs;
    return(OK);
```

New Code

b4ask.c

```
case BSIM4_MULT_I:
    value->rValue = here->BSIM4mult_i;
    return(OK);
case BSIM4_MULT_Q:
    value->rValue = here->BSIM4mult_q;
    return(OK);
case BSIM4_MULT_FN:
    value->rValue = here->BSIM4mult_fn;
    return(OK);

case BSIM4_SOURCECONDUCT:
    value->rValue = here->BSIM4sourceConductance;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_DRAINCONDUCT:
    value->rValue = here->BSIM4drainConductance;
    value->rValue *= here->BSIM4mult_i;
    return(OK);

case BSIM4_CD:
    value->rValue = here->BSIM4cd;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_CBS:
    value->rValue = here->BSIM4cbs;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_CBD:
    value->rValue = here->BSIM4cbd;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_CSUB:
    value->rValue = here->BSIM4csub;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_IGIDL:
    value->rValue = here->BSIM4Igidl;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
```

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```
case BSIM4_IGISL:
    value->rValue = here->BSIM4Igisl;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_IGS:
    value->rValue = here->BSIM4Igs;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_IGD:
    value->rValue = here->BSIM4Igd;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_IGB:
    value->rValue = here->BSIM4Igb;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_IGCS:
    value->rValue = here->BSIM4Igcs;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_IGCD:
    value->rValue = here->BSIM4Igcd;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_GM:
    value->rValue = here->BSIM4gm;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_GDS:
    value->rValue = here->BSIM4gds;
    value->rValue *= here->BSIM4mult_i;
    return(OK);

case BSIM4_GMBS:
    value->rValue = here->BSIM4gmbs;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_GBD:
    value->rValue = here->BSIM4gbd;
    value->rValue *= here->BSIM4mult_i;
    return(OK);
case BSIM4_GBS:
    value->rValue = here->BSIM4gbs;
    value->rValue *= here->BSIM4mult_i;
    return(OK);

case BSIM4_QB:
     value->rValue = here->BSIM4qbulk;
     value->rValue *= here->BSIM4mult_q;
     return(OK);
case BSIM4_QG:
     value->rValue = here->BSIM4qgate;
     value->rValue *= here->BSIM4mult_q;
     return(OK);
case BSIM4_QS:
     value->rValue = here->BSIM4qsrc;
     value->rValue *= here->BSIM4mult_q;
     return(OK);
case BSIM4_QD:
     value->rValue = here->BSIM4qdrn;
     value->rValue *= here->BSIM4mult_q;
     return(OK);
```

Summary of changes for BSIM4 4.8.3 release

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```
case BSIM4_GCRG:
    value->rValue = here->BSIM4gcrg;
    value->rValue *= here->BSIM4mult_i;
    return(OK);

case BSIM4_CGGB:
    value->rValue = here->BSIM4cggb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CGDB:
    value->rValue = here->BSIM4cgdb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CGSB:
    value->rValue = here->BSIM4cgsb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CDGB:
    value->rValue = here->BSIM4cdgb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CDDB:
    value->rValue = here->BSIM4cddb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CDSB:
    value->rValue = here->BSIM4cdsb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);

case BSIM4_CBGB:
    value->rValue = here->BSIM4cbgb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CBDB:
    value->rValue = here->BSIM4cbdb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CBSB:
    value->rValue = here->BSIM4cbsb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CSGB:
    value->rValue = here->BSIM4csgb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CSDB:
    value->rValue = here->BSIM4csdb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CSSB:
    value->rValue = here->BSIM4cssb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
```

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```
case BSIM4_CGBB:
    value->rValue = here->BSIM4cgbb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CDBB:
    value->rValue = here->BSIM4cdbb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CSBB:
    value->rValue = here->BSIM4csbb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CBBB:
    value->rValue = here->BSIM4cbbb;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CAPBD:
    value->rValue = here->BSIM4capbd;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
case BSIM4_CAPBS:
    value->rValue = here->BSIM4capbs;
    value->rValue *= here->BSIM4mult_q;
    return(OK);
```

New Code

bsim4def.h

```
double BSIM4mult_i;
double BSIM4mult_q;
double BSIM4mult_fn;

unsigned BSIM4mult_iGiven :1;
unsigned BSIM4mult_qGiven :1;
unsigned BSIM4mult_fnGiven :1;

#define BSIM4_MULT_I                    1280
#define BSIM4_MULT_Q                    1281
#define BSIM4_MULT_FN                   1282
```

New Code

b4.c

```
IOP( "mult_i", BSIM4_MULT_I, IF_REAL, "Variability in current"),
IOP( "mult_q", BSIM4_MULT_Q, IF_REAL, "Variability in charge"),
IOP( "mult_fn", BSIM4_MULT_FN, IF_REAL, "Variability in flicker noise"),
```

New Code

b4par.c

```
case BSIM4_MULT_I:
    here->BSIM4mult_i = value->rValue;
    here->BSIM4mult_iGiven = TRUE;
    break;
case BSIM4_MULT_Q:
    here->BSIM4mult_q = value->rValue;
    here->BSIM4mult_qGiven = TRUE;
    break;
case BSIM4_MULT_FN:
    here->BSIM4mult_fn = value->rValue;
    here->BSIM4mult_fnGiven = TRUE;
    break;
```

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New Code

b4set.c

```
if (!here->BSIM4mult_iGiven)
    here->BSIM4mult_i = 1.0;
if (!here->BSIM4mult_qGiven)
    here->BSIM4mult_q = 1.0;
if (!here->BSIM4mult_fnGiven)
    here->BSIM4mult_fn = here->BSIM4mult_i;
```

Note: For the following files mult_i, mult_q and mult_fn were multiplied at highlight places in the corresponding existing files.

New Code

b4ld.c

```
 mult_i = here->BSIM4mult_i;
 mult_q = here->BSIM4mult_q;

double mult_i, mult_q;

(*(ckt->CKTrhs + here->BSIM4dNodePrime) += (mult_i * (ceqjd - ceqbd + ceqgdtot
                                        - ceqdrn + Idtoteq) - mult_q * ceqqd));
(*(ckt->CKTrhs + here->BSIM4gNodePrime) -= mult_q * ceqqg - mult_i * (ceqgcrg - Igtoteq));
    (*(ckt->CKTrhs + here->BSIM4gNodeExt) -= mult_i * ceqgcrg);
(*(ckt->CKTrhs + here->BSIM4gNodeMid) -= mult_q * ceqqgmid + mult_i * ceqgcrg);
{   (*(ckt->CKTrhs + here->BSIM4bNodePrime) += (mult_i * (ceqbd + ceqbs - ceqjd
                                          - ceqjs + Ibtoteq) - mult_q * ceqqb));
    (*(ckt->CKTrhs + here->BSIM4sNodePrime) += (mult_i * (ceqdrn - ceqbs + ceqjs
                + ceqgstot + Istoteq) + mult_q * (ceqqg + ceqqb + ceqqd + ceqqgmid)));
{   (*(ckt->CKTrhs + here->BSIM4dbNode) -= mult_i * (ceqjd + ceqqjd));



(*(ckt->CKTrhs + here->BSIM4bNodePrime) += (mult_i * (ceqbd + ceqbs + Ibtoteq) - mult_q * ceqqb));
(*(ckt->CKTrhs + here->BSIM4sbNode) -= mult_i * (ceqjs + ceqqjs));
(*(ckt->CKTrhs + here->BSIM4sNodePrime) += (mult_i * (ceqdrn - ceqbs + ceqjs - ceqgstot + Istoteq) + mult_q * (ceqqd



{   (*(ckt->CKTrhs + here->BSIM4dNode) -= mult_i * ceqgdtot);
    (*(ckt->CKTrhs + here->BSIM4sNode) += mult_i * ceqgstot);
    {   (*(here->BSIM4GEgePtr) += mult_i * geltd);
        (*(here->BSIM4GPgePtr) -= mult_i * geltd);
        (*(here->BSIM4GEgpPtr) -= mult_i * geltd);
    (*(here->BSIM4GPgpPtr) += mult_q * (gcggb - ggtg) + mult_i * (geltd + gIgtotg));
    (*(here->BSIM4GPdpPtr) += mult_q * (gcgdb - ggtd) + mult_i * gIgtotd);
    (*(here->BSIM4GPspPtr) += mult_q * (gcgsb - ggts) + mult_i * gIgtots);
    (*(here->BSIM4GPbpPtr) += mult_q * (gcgbb - ggtb) + mult_i * gIgtotb);
{   (*(here->BSIM4GEgePtr) += mult_i * gcrg);
        (*(here->BSIM4GEgpPtr) += mult_i * gcrgg);
        (*(here->BSIM4GEdpPtr) += mult_i * gcrgd);
        (*(here->BSIM4GEspPtr) += mult_i * gcrgs);
    (*(here->BSIM4GEbpPtr) += mult_i * gcrgb);
        (*(here->BSIM4GPgePtr) -= mult_i * gcrg);
```

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```
    (*(here->BSIM4GPgpPtr) += mult_q * (gcggb - ggtg) + mult_i * (gIgtotg - gcrgg));
    (*(here->BSIM4GPdpPtr) += mult_q * (gcgdb - ggtd) + mult_i * (gIgtotd - gcrgd));
    (*(here->BSIM4GPspPtr) += mult_q * (gcgsb - ggts) + mult_i * (gIgtots - gcrgs));
    (*(here->BSIM4GPbpPtr) += mult_q * (gcgbb - ggtb) + mult_i * (gIgtotb - gcrgb));
{   (*(here->BSIM4GEgePtr) += mult_i * geltd);
        (*(here->BSIM4GEgmPtr) -= mult_i * geltd);
        (*(here->BSIM4GMgePtr) -= mult_i * geltd);
        (*(here->BSIM4GMgmPtr) += mult_i * (geltd + gcrg) + mult_q * gcgmgmb);
        (*(here->BSIM4GMdpPtr) += mult_i * gcrgd + mult_q * gcgmdb);
        (*(here->BSIM4GMgpPtr) += mult_i * gcrgg);
        (*(here->BSIM4GMspPtr) += mult_i * gcrgs + mult_q * gcgmsb);
        (*(here->BSIM4GMbpPtr) += mult_i * gcrgb + mult_q * gcgmbb);
        (*(here->BSIM4DPgmPtr) += mult_q * gcdgmb);
        (*(here->BSIM4GPgmPtr) -= mult_i * gcrg);
        (*(here->BSIM4SPgmPtr) += mult_q * gcsgmb);
        (*(here->BSIM4BPgmPtr) += mult_q * gcbgmb);
        (*(here->BSIM4GPgpPtr) += mult_q * (gcggb - ggtg) + mult_i * (gIgtotg - gcrgg));
        (*(here->BSIM4GPdpPtr) += mult_q * (gcgdb - ggtd) + mult_i * (gIgtotd - gcrgd));
        (*(here->BSIM4GPspPtr) += mult_q * (gcgsb - ggts) + mult_i * (gIgtots - gcrgs));
        (*(here->BSIM4GPbpPtr) += mult_q * (gcgbb - ggtb) + mult_i * (gIgtotb - gcrgb));

{   (*(here->BSIM4GPgpPtr) += mult_q * (gcggb - ggtg) + mult_i * gIgtotg);
    (*(here->BSIM4GPdpPtr) += mult_q * (gcgdb - ggtd) + mult_i * gIgtotd);
        (*(here->BSIM4GPspPtr) += mult_q * (gcgsb - ggts) + mult_i * gIgtots);
    (*(here->BSIM4GPbpPtr) += mult_q * (gcgbb - ggtb) + mult_i * gIgtotb);
{   (*(here->BSIM4DgpPtr) += mult_i * gdtotg);
    (*(here->BSIM4DspPtr) += mult_i * gdtots);
        (*(here->BSIM4DbpPtr) += mult_i * gdtotb);
        (*(here->BSIM4SdpPtr) += mult_i * gstotd);
        (*(here->BSIM4SgpPtr) += mult_i * gstotg);
        (*(here->BSIM4SbpPtr) += mult_i * gstotb);
    (*(here->BSIM4DPdpPtr) += mult_i * (gdpr + here->BSIM4gds + here->BSIM4gbd

                    - gdtotd + RevSum + gbdpdp - gIdtotd) + mult_q * (T1 * ddxpart_dVd + gcddb + dxpart * ggtd));
(*(here->BSIM4DPdPtr) -= mult_i * (gdpr + gdtot));
(*(here->BSIM4DPgpPtr) += mult_i * (Gm - gdtotg + gbdpg - gIdtotg)
       + mult_q * (dxpart * ggtg + T1 * ddxpart_dVg + gcdgb));
(*(here->BSIM4DPspPtr) -= mult_i * (here->BSIM4gds + gdtots + gIdtots
       + FwdSum - gbdpsp) - mult_q * (dxpart * ggts + T1 * ddxpart_dVs + gcdsb));
(*(here->BSIM4DPbpPtr) -= mult_i * (gjbd + gdtotb - Gmbs - gbdpb + gIdtotb)
       - mult_q * (dxpart * ggtb + gcdbb + T1 * ddxpart_dVb));
(*(here->BSIM4DdpPtr) -= mult_i * (gdpr - gdtotd));
(*(here->BSIM4DdPtr) += mult_i * (gdpr + gdtot));
(*(here->BSIM4SPdpPtr) -= mult_i * (here->BSIM4gds + gstotd + RevSum - gbspdp + gIstotd)
       - mult_q * (T1 * dsxpart_dVd + sxpart * ggtd + gcsdb));
(*(here->BSIM4SPgpPtr) += mult_q * (gcsgb + sxpart * ggtg + T1 * dsxpart_dVg) + mult_i * (gbspg - Gm - gstotg - gIstotg));
(*(here->BSIM4SPspPtr) += mult_i * (gspr + here->BSIM4gds + here->BSIM4gbs - gIstots - gstots + FwdSum + gbspsp) + mult_q * (sxpart
(*(here->BSIM4SPsPtr) -= mult_i * (gspr + gstot));
(*(here->BSIM4SPbpPtr) -= mult_i * (gjbs + gstotb + Gmbs - gbspb + gIstotb) - mult_q * (gcsbb + sxpart * ggtb + T1 * dsxpart_dVb));

(*(here->BSIM4SspPtr) -= mult_i * (gspr - gstots));
(*(here->BSIM4SsPtr) += mult_i * (gspr + gstot));
(*(here->BSIM4BPdpPtr) += mult_q * gcbdb - mult_i * (gjbd - gbbdp + gIbtotd));
(*(here->BSIM4BPgpPtr) += mult_q * gcbgb - mult_i * (here->BSIM4gbgs + gIbtotg));
(*(here->BSIM4BPspPtr) += mult_q * gcbsb - mult_i * (gjbs - gbbsp + gIbtots));
(*(here->BSIM4BPbpPtr) += mult_i * (gjbd + gjbs - here->BSIM4gbbs
        - gIbtotb) + mult_q * gcbbb);
(*(here->BSIM4DPdpPtr) += mult_i * ggidld);
(*(here->BSIM4DPgpPtr) += mult_i * ggidlg);
(*(here->BSIM4DPspPtr) -= mult_i * (ggidlg + ggidld + ggidlb));
(*(here->BSIM4DPbpPtr) += mult_i * ggidlb);
(*(here->BSIM4BPdpPtr) -= mult_i * ggidld);
(*(here->BSIM4BPgpPtr) -= mult_i * ggidlg);
(*(here->BSIM4BPspPtr) += mult_i * (ggidlg + ggidld + ggidlb));
(*(here->BSIM4BPbpPtr) -= mult_i * ggidlb);
(*(here->BSIM4SPdpPtr) -= mult_i * (ggisls + ggislg + ggislb));
(*(here->BSIM4SPgpPtr) += mult_i * ggislg);
(*(here->BSIM4SPspPtr) += mult_i * ggisls);
(*(here->BSIM4SPbpPtr) += mult_i * ggislb);
(*(here->BSIM4BPdpPtr) += mult_i * (ggislg + ggisls + ggislb));
(*(here->BSIM4BPgpPtr) -= mult_i * ggislg);
(*(here->BSIM4BPspPtr) -= mult_i * ggisls);
```

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```
(*(here->BSIM4BPbpPtr) -= mult_i * ggislb);
{   (*(here->BSIM4DPdbPtr) += mult_q * gcdbdb - mult_i * here->BSIM4gbd);
    (*(here->BSIM4SPsbPtr) -= mult_i * here->BSIM4gbs - mult_q * gcsbsb);
    (*(here->BSIM4DBdpPtr) += mult_q * gcdbdb - mult_i * here->BSIM4gbd);
    (*(here->BSIM4DBdbPtr) += mult_i * (here->BSIM4gbd + here->BSIM4grbpd + here->BSIM4grbdb) - mult_q * gcdbdb);

(*(here->BSIM4DBbpPtr) -= mult_i * here->BSIM4grbpd);
(*(here->BSIM4DBbPtr) -= mult_i * here->BSIM4grbdb);
(*(here->BSIM4BPdbPtr) -= mult_i * here->BSIM4grbpd);
(*(here->BSIM4BPbPtr) -= mult_i * here->BSIM4grbpb);
(*(here->BSIM4BPsbPtr) -= mult_i * here->BSIM4grbps);
(*(here->BSIM4BPbpPtr) += mult_i * (here->BSIM4grbpd + here->BSIM4grbps
(*(here->BSIM4SBspPtr) += mult_q * gcsbsb - mult_i * here->BSIM4gbs);
(*(here->BSIM4SBbpPtr) -= mult_i * here->BSIM4grbps);
(*(here->BSIM4SBbPtr) -= mult_i * here->BSIM4grbsb);

(*(here->BSIM4SBsbPtr) += mult_i * (here->BSIM4gbs + here->BSIM4grbps + here->BSIM4grbsb) - mult_q * gcsbsb);

(*(here->BSIM4BdbPtr) -= mult_i * here->BSIM4grbdb);
(*(here->BSIM4BbpPtr) -= mult_i * here->BSIM4grbpb);
(*(here->BSIM4BsbPtr) -= mult_i * here->BSIM4grbsb);
(*(here->BSIM4BbPtr) += mult_i * (here->BSIM4grbsb + here->BSIM4grbdb
```

New Code

b4noi.c

```
double mult_i, mult_fn;

mult_i = here->BSIM4mult_i;
mult_fn = here->BSIM4mult_fn;
         gdpr * mult_i);
         gspr * mult_i);
                        here->BSIM4grgeltd * mult_i);
                        here->BSIM4grgeltd * mult_i/T1);
                        here->BSIM4grgeltd * mult_i);
           here->BSIM4grbps * mult_i);
           here->BSIM4grbpd * mult_i);
           here->BSIM4grbpb * mult_i);
           here->BSIM4grbsb * mult_i);
           here->BSIM4grbdb * mult_i);
           here->BSIM4grbps * mult_i);
           here->BSIM4grbpd * mult_i);
           here->BSIM4grbpb * mult_i);
           here->BSIM4grbpb * mult_i);
             (T0 / T1) * model->BSIM4ntnoi * mult_i);
                  (T0 / T1) * model->BSIM4ntnoi * mult_i);
                                here->BSIM4sNodePrime, (T2 - igsquare) * mult_i);
                       here->BSIM4sNodePrime, (T2 - igsquare) * mult_i);
                                here->BSIM4sNodePrime, T2 * T4 * mult_i);
                                here->BSIM4sNodePrime, T2 * T3 * mult_i,
            T2 * T7 * mult_i, 0.5 * M_PI);
                                here->BSIM4dNodePrime, T2 * T3 * mult_i,
            T2 * T7 * mult_i, 0.5 * M_PI);
           noizDens[BSIM4FLNOIZ] *= mult_fn * model->BSIM4kf
                             noizDens[BSIM4FLNOIZ] *= mult_fn * (Ssi * Swi) / T1;
                      mult_i * (here->BSIM4Igs + here->BSIM4Igcs));
                      mult_i * (here->BSIM4Igd + here->BSIM4Igcd));
                      mult_i * (here->BSIM4Igs + here->BSIM4Igcd));
                      mult_i * (here->BSIM4Igd + here->BSIM4Igcs));
                      mult i * here->BSIM4Igb);
```

New Code

b4acld.c

```
double mult_i, mult_q;


mult_i = here->BSIM4mult_i;
mult_q = here->BSIM4mult_q;
```

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```
    {   *(here->BSIM4GEgePtr) += mult_i * geltd;
        *(here->BSIM4GPgePtr) -= mult_i * geltd;
        *(here->BSIM4GEgpPtr) -= mult_i * geltd;
        *(here->BSIM4GPgpPtr +1) += mult_q * xcggbr;
*(here->BSIM4GPgpPtr) += mult_i * (geltd + gIgtotg) + mult_q * xcggbi;
        *(here->BSIM4GPdpPtr +1) += mult_q * xcgdbr;
        *(here->BSIM4GPdpPtr) += mult_q * xcgdbi + mult_i * gIgtotd;
        *(here->BSIM4GPspPtr +1) += mult_q * xcgsbr;
        *(here->BSIM4GPspPtr) += mult_q * xcgsbi + mult_i * gIgtots;
        *(here->BSIM4GPbpPtr +1) += mult_q * xcgbbr;
        *(here->BSIM4GPbpPtr) += mult_q * xcgbbi + mult_i * gIgtotb;
    {   *(here->BSIM4GEgePtr) += mult_i * gcrg;
        *(here->BSIM4GEgpPtr) += mult_i * gcrgg;
        *(here->BSIM4GEdpPtr) += mult_i * gcrgd;
        *(here->BSIM4GEspPtr) += mult_i * gcrgs;
        *(here->BSIM4GEbpPtr) += mult_i * gcrgb;
        *(here->BSIM4GPgePtr) -= mult_i * gcrg;
        *(here->BSIM4GPgpPtr +1) += mult_q * xcggbr;
*(here->BSIM4GPgpPtr) -= mult_i * (gcrgg - gIgtotg) - mult_q * xcggbi;
        *(here->BSIM4GPdpPtr +1) += mult_q * xcgdbr;
*(here->BSIM4GPdpPtr) -= mult_i * (gcrgd - gIgtotd) - mult_q * xcgdbi;
        *(here->BSIM4GPspPtr +1) += mult_q * xcgsbr;
*(here->BSIM4GPspPtr) -= mult_i * (gcrgs - gIgtots) - mult_q * xcgsbi;
        *(here->BSIM4GPbpPtr +1) += mult_q * xcgbbr;
*(here->BSIM4GPbpPtr) -= mult_i * (gcrgb - gIgtotb) - mult_q * xcgbbi;

{   *(here->BSIM4GEgePtr) += mult_i * geltd;
    *(here->BSIM4GEgmPtr) -= mult_i * geltd;
    *(here->BSIM4GMgePtr) -= mult_i * geltd;
    *(here->BSIM4GMgmPtr) += mult_i * (geltd + gcrg);
    *(here->BSIM4GMgmPtr +1) += mult_q * xcgmgmb;
    *(here->BSIM4GMdpPtr) += mult_i * gcrgd;
    *(here->BSIM4GMdpPtr +1) += mult_q * xcgmdb;
    *(here->BSIM4GMgpPtr) += mult_i * gcrgg;
    *(here->BSIM4GMspPtr) += mult_i * gcrgs;
    *(here->BSIM4GMspPtr +1) += mult_q * xcgmsb;
    *(here->BSIM4GMbpPtr) += mult_i * gcrgb;
    *(here->BSIM4GMbpPtr +1) += mult_q * xcgmbb;
    *(here->BSIM4DPgmPtr +1) += mult_q * xcdgmb;
    *(here->BSIM4GPgmPtr) -= mult_i * gcrg;
    *(here->BSIM4SPgmPtr +1) += mult_q * xcsgmb;
    *(here->BSIM4BPgmPtr +1) += mult_q * xcbgmb;
    *(here->BSIM4GPgpPtr) -= mult_i * (gcrgg - gIgtotg) - mult_q * xcggbi;
    *(here->BSIM4GPgpPtr +1) += mult_q * xcggbr;
    *(here->BSIM4GPdpPtr) -= mult_i * (gcrgd - gIgtotd) - mult_q * xcgdbi;
    *(here->BSIM4GPdpPtr +1) += mult_q * xcgdbr;
    *(here->BSIM4GPspPtr) -= mult_i * (gcrgs - gIgtots) - mult_q * xcgsbi;
    *(here->BSIM4GPspPtr +1) += mult_q * xcgsbr;
    *(here->BSIM4GPbpPtr) -= mult_i * (gcrgb - gIgtotb) - mult_q * xcgbbi;
    *(here->BSIM4GPbpPtr +1) += mult_q * xcgbbr;
```

```
{   *(here->BSIM4GPgpPtr +1) += mult_q * xcggbr;
    *(here->BSIM4GPgpPtr) += mult_q * xcggbi + mult_i * gIgtotg;
    *(here->BSIM4GPdpPtr +1) += mult_q * xcgdbr;
    *(here->BSIM4GPdpPtr) += mult_q * xcgdbi + mult_i * gIgtotd;
    *(here->BSIM4GPspPtr +1) += mult_q * xcgsbr;
    *(here->BSIM4GPspPtr) += mult_q * xcgsbi + mult_i * gIgtots;
    *(here->BSIM4GPbpPtr +1) += mult_q * xcgbbr;
    *(here->BSIM4GPbpPtr) += mult_q * xcgbbi + mult_i * gIgtotb;
{   (*(here->BSIM4DgpPtr) += mult_i * gdtotg);
    (*(here->BSIM4DspPtr) += mult_i * gdtots);
    (*(here->BSIM4DbpPtr) += mult_i * gdtotb);
    (*(here->BSIM4SdpPtr) += mult_i * gstotd);
    (*(here->BSIM4SgpPtr) += mult_i * gstotg);
    (*(here->BSIM4SbpPtr) += mult_i * gstotb);
*(here->BSIM4DPdpPtr +1) += mult_q * xcddbr + mult_i * (gdsi + RevSumi);
*(here->BSIM4DPdpPtr) += + mult_i * (gdpr + gdsr + here->BSIM4gbd
       - gdtotd + RevSumr + gbdpdp - gIdtotd) + mult_q * xcddbi;
*(here->BSIM4DPdPtr) -= mult_i * (gdpr + gdtot);
*(here->BSIM4DPgpPtr +1) += mult_q * xcdgbr + mult_i * Gmi;
*(here->BSIM4DPgpPtr) += mult_i * (Gmr - gdtotg + gbdpg - gIdtotg) + mult_q * xcdgbi;
*(here->BSIM4DPspPtr +1) += mult_q * xcdsbr - mult_i * (gdsi + FwdSumi);
*(here->BSIM4DPspPtr) -= mult_i * (gdsr + FwdSumr + gdtots - gbdpsp + gIdtots) - mult_q * xcdsbi;
*(here->BSIM4DPbpPtr +1) += mult_q * xcdbbr + mult_i * Gmbsi;
*(here->BSIM4DPbpPtr) -= mult_i * (gjbd + gdtotb - Gmbsr - gbdpb + gIdtotb) - mult_q * xcdbbi;
*(here->BSIM4DdpPtr) -= mult_i * (gdpr - gdtotd);
*(here->BSIM4DdPtr) += mult_i * (gdpr + gdtot);
*(here->BSIM4SPdpPtr +1) += mult_q * xcsdbr - mult_i * (gdsi + RevSumi);
*(here->BSIM4SPdpPtr) -= mult_i * (gdsr + gstotd + RevSumr - gbspdp + gIstotd) - mult_q * xcsdbi;
*(here->BSIM4SPgpPtr +1) += mult_q * xcsgbr - mult_i * Gmi;
*(here->BSIM4SPgpPtr) -= mult_i * (Gmr + gstotg - gbspg + gIstotg) - mult_q * xcsgbi;

*(here->BSIM4SPspPtr +1) += mult_q * xcssbr + mult_i * (gdsi + FwdSumi);
*(here->BSIM4SPspPtr) += mult_i * (gspr + gdsr + here->BSIM4gbs
       - gstots + FwdSumr + gbspsp - gIstots) + mult_q * xcssbi;
*(here->BSIM4SPsPtr) -= mult_i * (gspr + gstot);
*(here->BSIM4SPbpPtr +1) += mult_q * xcsbbr - mult_i * Gmbsi;
*(here->BSIM4SPbpPtr) -= mult_i * (gjbs + gstotb + Gmbsr - gbspb + gIstotb) - mult_q * xcsbbi;
*(here->BSIM4SspPtr) -= mult_i * (gspr - gstots);
*(here->BSIM4SsPtr) += mult_i * (gspr + gstot);
*(here->BSIM4BPdpPtr +1) += mult_q * xcbdb;
*(here->BSIM4BPdpPtr) -= mult_i * (gjbd - gbbdp + gIbtotd);
*(here->BSIM4BPgpPtr +1) += mult_q * xcbgb;
*(here->BSIM4BPgpPtr) -= mult_i * (here->BSIM4gbgs + gIbtotg);
*(here->BSIM4BPspPtr +1) += mult_q * xcbsb;
*(here->BSIM4BPspPtr) -= mult_i * (gjbs - gbbsp + gIbtots);
*(here->BSIM4BPbpPtr +1) += mult_q * xcbbb;

   *(here->BSIM4BPbpPtr) += mult_i * (gjbd + gjbs - here->BSIM4gbbs
(*(here->BSIM4DPdpPtr) += mult_i * ggidld);
(*(here->BSIM4DPgpPtr) += mult_i * ggidlg);
(*(here->BSIM4DPspPtr) -= mult_i * ((ggidlg + ggidld) + ggidlb));
(*(here->BSIM4DPbpPtr) += mult_i * ggidlb);
(*(here->BSIM4BPdpPtr) -= mult_i * ggidld);
(*(here->BSIM4BPgpPtr) -= mult_i * ggidlg);
(*(here->BSIM4BPspPtr) += mult_i * ((ggidlg + ggidld) + ggidlb));
(*(here->BSIM4BPbpPtr) -= mult_i * ggidlb);
(*(here->BSIM4SPdpPtr) -= mult_i * ((ggisls + ggislg) + ggislb));
(*(here->BSIM4SPgpPtr) += mult_i * ggislg);
(*(here->BSIM4SPspPtr) += mult_i * ggisls);
(*(here->BSIM4SPbpPtr) += mult_i * ggislb);
(*(here->BSIM4BPdpPtr) += mult_i * ((ggislg + ggisls) + ggislb));
(*(here->BSIM4BPgpPtr) -= mult_i * ggislg);
(*(here->BSIM4BPspPtr) -= mult_i * ggisls);
(*(here->BSIM4BPbpPtr) -= mult_i * ggislb);
```

Summary of changes for BSIM4 4.8.3 release

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```
{   (*(here->BSIM4DPdbPtr +1) += mult_q * xcdbdb);
    (*(here->BSIM4DPdbPtr) -= mult_i * here->BSIM4gbd);
    (*(here->BSIM4SPsbPtr +1) += mult_q * xcsbsb);
    (*(here->BSIM4SPsbPtr) -= mult_i * here->BSIM4gbs);
    (*(here->BSIM4DBdpPtr +1) += mult_q * xcdbdb);
    (*(here->BSIM4DBdpPtr) -= mult_i * here->BSIM4gbd);
    (*(here->BSIM4DBdbPtr +1) -= mult_q * xcdbdb);
    (*(here->BSIM4DBdbPtr) += mult_i * (here->BSIM4gbd + here->BSIM4grbpd
    (*(here->BSIM4DBbpPtr) -= mult_i * here->BSIM4grbpd);
    (*(here->BSIM4DBbPtr) -= mult_i * here->BSIM4grbdb);
    (*(here->BSIM4BPdbPtr) -= mult_i * here->BSIM4grbpd);
    (*(here->BSIM4BPbPtr) -= mult_i * here->BSIM4grbpb);
    (*(here->BSIM4BPsbPtr) -= mult_i * here->BSIM4grbps);
    (*(here->BSIM4BPbpPtr) += mult_i * (here->BSIM4grbpd + here->BSIM4grbps
    (*(here->BSIM4SBspPtr +1) += mult_q * xcsbsb);
    (*(here->BSIM4SBspPtr) -= mult_i * here->BSIM4gbs);
    (*(here->BSIM4SBbpPtr) -= mult_i * here->BSIM4grbps);
    (*(here->BSIM4SBbPtr) -= mult_i * here->BSIM4grbsb);
    (*(here->BSIM4SBsbPtr +1) -= mult_q * xcsbsb);
    (*(here->BSIM4SBsbPtr) += mult_i * (here->BSIM4gbs
    (*(here->BSIM4BdbPtr) -= mult_i * here->BSIM4grbdb);
    (*(here->BSIM4BbpPtr) -= mult_i * here->BSIM4grbpb);
    (*(here->BSIM4BsbPtr) -= mult_i * here->BSIM4grbsb);
    (*(here->BSIM4BbPtr) += mult_i * (here->BSIM4grbsb + here->BSIM4grbdb
```

New Code

b4pzld.c

```
double mult_i, mult_q;



 mult_i = here->BSIM4mult_i;
 mult_q = here->BSIM4mult_q;

     {   *(here->BSIM4GEgePtr) += mult_i * geltd;
         *(here->BSIM4GPgePtr) -= mult_i * geltd;
         *(here->BSIM4GEgpPtr) -= mult_i * geltd;
         *(here->BSIM4GPgpPtr ) += mult_q * xcggb * s->real;
         *(here->BSIM4GPgpPtr +1) += mult_q * xcggb * s->imag;
         *(here->BSIM4GPgpPtr) += mult_i * (geltd + gIgtotg) - mult_q * xgtg;
         *(here->BSIM4GPdpPtr ) += mult_q * xcgdb * s->real;
         *(here->BSIM4GPdpPtr +1) += mult_q * xcgdb * s->imag;
*(here->BSIM4GPdpPtr) -= mult_q * xgtd - mult_i * gIgtotd;
         *(here->BSIM4GPspPtr ) += mult_q * xcgsb * s->real;
         *(here->BSIM4GPspPtr +1) += mult_q * xcgsb * s->imag;
         *(here->BSIM4GPspPtr) -= mult_q * xgts - mult_i * gIgtots;
         *(here->BSIM4GPbpPtr ) += mult_q * xcgbb * s->real;
         *(here->BSIM4GPbpPtr +1) += mult_q * xcgbb * s->imag;
*(here->BSIM4GPbpPtr) -= mult_q * xgtb - mult_i * gIgtotb;
     {   *(here->BSIM4GEgePtr) += mult_i * gcrg;
         *(here->BSIM4GEgpPtr) += mult_i * gcrgg;
         *(here->BSIM4GEdpPtr) += mult_i * gcrgd;
         *(here->BSIM4GEspPtr) += mult_i * gcrgs;
         *(here->BSIM4GEbpPtr) += mult_i * gcrgb;
         *(here->BSIM4GPgePtr) -= mult_i * gcrg;
         *(here->BSIM4GPgpPtr ) += mult_q * xcggb * s->real;
         *(here->BSIM4GPgpPtr +1) += mult_q * xcggb * s->imag;
         *(here->BSIM4GPgpPtr) -= mult_i * (gcrgg - gIgtotg) + mult_q * xgtg;
         *(here->BSIM4GPdpPtr ) += mult_q * xcgdb * s->real;
         *(here->BSIM4GPdpPtr +1) += mult_q * xcgdb * s->imag;
         *(here->BSIM4GPdpPtr) -= mult_i * (gcrgd - gIgtotd) + mult_q * xgtd;
         *(here->BSIM4GPspPtr ) += mult_q * xcgsb * s->real;
         *(here->BSIM4GPspPtr +1) += mult_q * xcgsb * s->imag;
```

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```
    *(here->BSIM4GPspPtr) -= mult_i * (gcrgs - gIgtots) + mult_q * xgts;
    *(here->BSIM4GPbpPtr ) += mult_q * xcgbb * s->real;
    *(here->BSIM4GPbpPtr +1) += mult_q * xcgbb * s->imag;
    *(here->BSIM4GPbpPtr) -= mult_i * (gcrgb - gIgtotb) + mult_q * xgtb;
{   *(here->BSIM4GEgePtr) += mult_i * geltd;
    *(here->BSIM4GEgmPtr) -= mult_i * geltd;
    *(here->BSIM4GMgePtr) -= mult_i * geltd;
    *(here->BSIM4GMgmPtr) += mult_i * (geltd + gcrg);
    *(here->BSIM4GMgmPtr ) += mult_q * xcgmgmb * s->real;
    *(here->BSIM4GMgmPtr +1) += mult_q * xcgmgmb * s->imag;
    *(here->BSIM4GMdpPtr) += mult_i * gcrgd;
    *(here->BSIM4GMdpPtr ) += mult_q * xcgmdb * s->real;
    *(here->BSIM4GMdpPtr +1) += mult_q * xcgmdb * s->imag;
    *(here->BSIM4GMgpPtr) += mult_i * gcrgg;
    *(here->BSIM4GMspPtr) += mult_i * gcrgs;
    *(here->BSIM4GMspPtr ) += mult_q * xcgmsb * s->real;
    *(here->BSIM4GMspPtr +1) += mult_q * xcgmsb * s->imag;
    *(here->BSIM4GMbpPtr) += mult_i * gcrgb;
    *(here->BSIM4GMbpPtr ) += mult_q * xcgmbb * s->real;
    *(here->BSIM4GMbpPtr +1) += mult_q * xcgmbb * s->imag;
    *(here->BSIM4DPgmPtr ) += mult_q * xcdgmb * s->real;
    *(here->BSIM4DPgmPtr +1) += mult_q * xcdgmb * s->imag;
    *(here->BSIM4GPgmPtr) -= mult_i * gcrg;
    *(here->BSIM4SPgmPtr ) += mult_q * xcsgmb * s->real;
    *(here->BSIM4SPgmPtr +1) += mult_q * xcsgmb * s->imag;
    *(here->BSIM4BPgmPtr ) += mult_q * xcbgmb * s->real;
    *(here->BSIM4BPgmPtr +1) += mult_q * xcbgmb * s->imag;
    *(here->BSIM4GPgpPtr) -= mult_i * (gcrgg - gIgtotg) + mult_q * xgtg;

        *(here->BSIM4GPgpPtr ) += mult_q * xcggb * s->real;
        *(here->BSIM4GPgpPtr +1) += mult_q * xcggb * s->imag;
        *(here->BSIM4GPdpPtr) -= mult_i * (gcrgd - gIgtotd) + mult_q * xgtd;
        *(here->BSIM4GPdpPtr ) += mult_q * xcgdb * s->real;
        *(here->BSIM4GPdpPtr +1) += mult_q * xcgdb * s->imag;
        *(here->BSIM4GPspPtr) -= mult_i * (gcrgs - gIgtots) + mult_q * xgts;
        *(here->BSIM4GPspPtr ) += mult_q * xcgsb * s->real;
        *(here->BSIM4GPspPtr +1) += mult_q * xcgsb * s->imag;
        *(here->BSIM4GPbpPtr) -= mult_i * (gcrgb - gIgtotb) + mult_q * xgtb;
        *(here->BSIM4GPbpPtr ) += mult_q * xcgbb * s->real;
        *(here->BSIM4GPbpPtr +1) += mult_q * xcgbb * s->imag;
    {   *(here->BSIM4GPdpPtr ) += mult_q * xcgdb * s->real;
        *(here->BSIM4GPdpPtr +1) += mult_q * xcgdb * s->imag;
*(here->BSIM4GPdpPtr) -= mult_q * xgtd - mult_i * gIgtotd;
        *(here->BSIM4GPgpPtr ) += mult_q * xcggb * s->real;
        *(here->BSIM4GPgpPtr +1) += mult_q * xcggb * s->imag;
*(here->BSIM4GPgpPtr) -= mult_q * xgtg - mult_i * gIgtotg;
        *(here->BSIM4GPspPtr ) += mult_q * xcgsb * s->real;
        *(here->BSIM4GPspPtr +1) += mult_q * xcgsb * s->imag;
        *(here->BSIM4GPspPtr) -= mult_q * xgts - mult_i * gIgtots;
        *(here->BSIM4GPbpPtr ) += mult_q * xcgbb * s->real;
        *(here->BSIM4GPbpPtr +1) += mult_q * xcgbb * s->imag;
*(here->BSIM4GPbpPtr) -= mult_q * xgtb - mult_i * gIgtotb;
    {   (*(here->BSIM4DgpPtr) += mult_i * gdtotg);
        (*(here->BSIM4DspPtr) += mult_i * gdtots);
        (*(here->BSIM4DbpPtr) += mult_i * gdtotb);
        (*(here->BSIM4SdpPtr) += mult_i * gstotd);
        (*(here->BSIM4SgpPtr) += mult_i * gstotg);
        (*(here->BSIM4SbpPtr) += mult_i * gstotb);
```

Summary of changes for BSIM4 4.8.3 release

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```
*(here->BSIM4DPdpPtr ) += mult_q * xcddb * s->real;
*(here->BSIM4DPdpPtr +1) += mult_q * xcddb * s->imag;
*(here->BSIM4DPdpPtr) += mult_i * (gdpr + gds + here->BSIM4gbd
       + mult_q * (dxpart * xgtd + T1 * ddxpart_dVd);
*(here->BSIM4DPdPtr) -= mult_i * (gdpr + gdtot);
*(here->BSIM4DPgpPtr ) += mult_q * xcdgb * s->real;
*(here->BSIM4DPgpPtr +1) += mult_q * xcdgb * s->imag;
*(here->BSIM4DPgpPtr) += mult_i * (Gm - gdtotg + gbdpg - gIdtotg)
       + mult_q * (T1 * ddxpart_dVg + dxpart * xgtg);
*(here->BSIM4DPspPtr ) += mult_q * xcdsb * s->real;
*(here->BSIM4DPspPtr +1) += mult_q * xcdsb * s->imag;
*(here->BSIM4DPspPtr) -= mult_i * (gds + FwdSum + gdtots - gbdpsp + gIdtots)
       - mult_q * (T1 * ddxpart_dVs + dxpart * xgts);
*(here->BSIM4DPbpPtr ) += mult_q * xcdbb * s->real;
*(here->BSIM4DPbpPtr +1) += mult_q * xcdbb * s->imag;
*(here->BSIM4DPbpPtr) -= mult_i * (gjbd + gdtotb - Gmbs - gbdpb + gIdtotb)
       - mult_q * T1 * (ddxpart_dVb + dxpart * xgtb);
*(here->BSIM4DdpPtr) -= mult_i * (gdpr - gdtotd);
*(here->BSIM4DdPtr) += mult_i * (gdpr + gdtot);
*(here->BSIM4SPdpPtr ) += mult_q * xcsdb * s->real;
*(here->BSIM4SPdpPtr +1) += mult_q * xcsdb * s->imag;
*(here->BSIM4SPdpPtr) -= mult_i * (gds + gstotd + RevSum - gbspdp + gIstotd)
       - mult_q * (T1 * dsxpart_dVd + sxpart * xgtd);
*(here->BSIM4SPgpPtr ) += mult_q * xcsgb * s->real;
*(here->BSIM4SPgpPtr +1) += mult_q * xcsgb * s->imag;
*(here->BSIM4SPgpPtr) -= mult_i * (Gm + gstotg - gbspg + gIstotg)
       - mult_q * (T1 * dsxpart_dVg + sxpart * xgtg);
*(here->BSIM4SPspPtr ) += mult_q * xcssb * s->real;
*(here->BSIM4SPspPtr +1) += mult_q * xcssb * s->imag;
*(here->BSIM4SPspPtr) += mult_i * (gspr + gds + here->BSIM4gbs - gIstots
       + mult_q * (sxpart * xgts + T1 * dsxpart_dVs);
*(here->BSIM4SPsPtr) -= mult_i * (gspr + gstot);
*(here->BSIM4SPbpPtr ) += mult_q * xcsbb * s->real;
*(here->BSIM4SPbpPtr +1) += mult_q * xcsbb * s->imag;

   *(here->BSIM4SPbpPtr) -= mult_i * (gjbs + gstotb + Gmbs - gbspb + gIstotb)
          - mult_q * (T1 * dsxpart_dVb + sxpart * xgtb);
   *(here->BSIM4SspPtr) -= mult_i * (gspr - gstots);
   *(here->BSIM4SsPtr) += mult_i * (gspr + gstot);
   *(here->BSIM4BPdpPtr ) += mult_q * xcbdb * s->real;
   *(here->BSIM4BPdpPtr +1) += mult_q * xcbdb * s->imag;
   *(here->BSIM4BPdpPtr) -= mult_i * (gjbd - gbbdp + gIbtotd);
   *(here->BSIM4BPgpPtr ) += mult_q * xcbgb * s->real;
   *(here->BSIM4BPgpPtr +1) += mult_q * xcbgb * s->imag;
   *(here->BSIM4BPgpPtr) -= mult_i * (here->BSIM4gbgs + gIbtotg);
   *(here->BSIM4BPspPtr ) += mult_q * xcbsb * s->real;
   *(here->BSIM4BPspPtr +1) += mult_q * xcbsb * s->imag;
   *(here->BSIM4BPspPtr) -= mult_i * (gjbs - gbbsp + gIbtots);
   *(here->BSIM4BPbpPtr ) += mult_q * xcbbb * s->real;
   *(here->BSIM4BPbpPtr +1) += mult_q * xcbbb * s->imag;
   *(here->BSIM4BPbpPtr) += mult_i * (gjbd + gjbs - here->BSIM4gbbs
(*(here->BSIM4DPdpPtr) += mult_i * ggidld);
(*(here->BSIM4DPgpPtr) += mult_i * ggidlg);
(*(here->BSIM4DPspPtr) -= mult_i * ((ggidlg + ggidld) + ggidlb));
(*(here->BSIM4DPbpPtr) += mult_i * ggidlb);
(*(here->BSIM4BPdpPtr) -= mult_i * ggidld);
(*(here->BSIM4BPgpPtr) -= mult_i * ggidlg);
(*(here->BSIM4BPspPtr) += mult_i * ((ggidlg + ggidld) + ggidlb));
```

```
(*(here->BSIM4BPbpPtr) -= mult_i * ggidlb);
(*(here->BSIM4SPdpPtr) -= mult_i * ((ggisls + ggislg) + ggislb));
(*(here->BSIM4SPgpPtr) += mult_i * ggislg);
(*(here->BSIM4SPspPtr) += mult_i * ggisls);
(*(here->BSIM4SPbpPtr) += mult_i * ggislb);
(*(here->BSIM4BPdpPtr) += mult_i * ((ggislg + ggisls) + ggislb));
(*(here->BSIM4BPgpPtr) -= mult_i * ggislg);
(*(here->BSIM4BPspPtr) -= mult_i * ggisls);
(*(here->BSIM4BPbpPtr) -= mult_i * ggislb);
   {    (*(here->BSIM4DPdbPtr ) += mult_q * xcdbdb * s->real);
        (*(here->BSIM4DPdbPtr +1) += mult_q * xcdbdb * s->imag);
        (*(here->BSIM4DPdbPtr) -= mult_i * here->BSIM4gbd);
        (*(here->BSIM4SPsbPtr ) += mult_q * xcsbsb * s->real);
        (*(here->BSIM4SPsbPtr +1) += mult_q * xcsbsb * s->imag);
        (*(here->BSIM4SPsbPtr) -= mult_i * here->BSIM4gbs);
        (*(here->BSIM4DBdpPtr ) += mult_q * xcdbdb * s->real);
        (*(here->BSIM4DBdpPtr +1) += mult_q * xcdbdb * s->imag);
        (*(here->BSIM4DBdpPtr) -= mult_i * here->BSIM4gbd);
        (*(here->BSIM4DBdbPtr ) -= mult_q * xcdbdb * s->real);
        (*(here->BSIM4DBdbPtr +1) -= mult_q * xcdbdb * s->imag);
        (*(here->BSIM4DBdbPtr) += mult_i * (here->BSIM4gbd + here->BSIM4grbpd
        (*(here->BSIM4DBbpPtr) -= mult_i * here->BSIM4grbpd);
        (*(here->BSIM4DBbPtr) -= mult_i * here->BSIM4grbdb);
        (*(here->BSIM4BPdbPtr) -= mult_i * here->BSIM4grbpd);
        (*(here->BSIM4BPbPtr) -= mult_i * here->BSIM4grbpb);
        (*(here->BSIM4BPsbPtr) -= mult_i * here->BSIM4grbps);
        (*(here->BSIM4BPbpPtr) += mult_i * (here->BSIM4grbpd + here->BSIM4grbps
        (*(here->BSIM4SBspPtr ) += mult_q * xcsbsb * s->real);
        (*(here->BSIM4SBspPtr +1) += mult_q * xcsbsb * s->imag);
        (*(here->BSIM4SBspPtr) -= mult_i * here->BSIM4gbs);
        (*(here->BSIM4SBbpPtr) -= mult_i * here->BSIM4grbps);
        (*(here->BSIM4SBbPtr) -= mult_i * here->BSIM4grbsb);
        (*(here->BSIM4SBsbPtr ) -= mult_q * xcsbsb * s->real);
        (*(here->BSIM4SBsbPtr +1) -= mult_q * xcsbsb * s->imag);
        (*(here->BSIM4SBsbPtr) += mult_i * (here->BSIM4gbs
        (*(here->BSIM4BdbPtr) -= mult_i * here->BSIM4grbdb);
        (*(here->BSIM4BbpPtr) -= mult_i * here->BSIM4grbpb);
        (*(here->BSIM4BsbPtr) -= mult_i * here->BSIM4grbsb);
        (*(here->BSIM4BbPtr) += mult_i * (here->BSIM4grbsb + here->BSIM4grbdb
```

10) 2023bug1: Inclusion of comprehensive QA test for MULT check (Cadence/BSIM group)

To test the MULT implementations 7 new QA tests were added (Test no. 32 to 38).

For the case of nmos, they are as follows:

```
test                    32_DC_WL10
biases                  V(s)=0 V(b)=0
biasList                V(g)=0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2
biasSweep               V(d)=0.1,1.2,0.1
outputs                 I(d)
instanceParameters      w=1.2e-6 l=1e-6
modelParameters         parameters/nmosParameters
modelParameters         rbodymod=1 rgatemod=2 rdsmod=1
```

DC_Mode setup

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List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

```
289  test                    33_DC_multi
290  biases                  V(s)=0 V(b)=0
291  biasList                V(g)=0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2
292  biasSweep               V(d)=0.1,1.2,0.1
293  outputs                 I(d)
294  instanceParameters      w=1.2e-6 l=1e-6 mult_i=2
295  modelParameters         parameters/nmosParameters
296  modelParameters         rbodymod=1 rgatemod=2 rdsmod=1
```

MULT_I

```
298  test                    34_DC_vb_multi
299  biases                  V(s)=0 V(b)=-0.2
300  biasList                V(g)=0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2
301  biasSweep               V(d)=0.1,1.2,0.1
302  outputs                 I(d)
303  instanceParameters      w=1.2e-6 l=1e-6 mult_i=2
304  modelParameters         parameters/nmosParameters
305  modelParameters         rbodymod=1 rgatemod=2 rdsmod=1
```

Mult_I back bias

```
307  test                    35_AC_Vd_MOD
308  temperature             27 150
309  biases                  V(s)=0 V(b)=0 V(g)=1.2
310  biasSweep               V(d)=0.1,1.2,0.1
311  outputs                 C(g,g) C(g,s) C(g,d)
312  instanceParameters      w=1.2e-6 l=1e-6
313  modelParameters         parameters/nmosParameters
314  modelParameters         rbodymod=1 rgatemod=2 rdsmod=1
```

AC – Mode setup

```
316  test                    36_AC_Vd_multq
317  temperature             27
318  biases                  V(s)=0 V(b)=0 V(g)=1.2
319  biasSweep               V(d)=0.1,1.2,0.1
320  outputs                 C(g,g) C(g,s) C(g,d)
321  instanceParameters      w=1.2e-6 l=1e-6 mult_q=2
322  modelParameters         parameters/nmosParameters
323  modelParameters         rbodymod=1 rgatemod=2 rdsmod=1 ketac=-2 keta=0
```

MULT_Q

```
325  test                    37_Noise5
326  temperature             27
327  biases                  V(s)=0 V(b)=0 V(d)=1.0
328  biasList                V(g)=0.6,0.8,1.0
329  freq                    dec 10 1e3 1e8
330  outputs                 N(d)
331  instanceParameters      w=1.2e-6 l=1e-6
332  modelParameters         parameters/nmosParameters
333  modelParameters         rbodymod=1 rgatemod=2 rdsmod=1
```

Noise – Mode setup

```
335  test                    38_Noise5_multfn
336  temperature             27
337  biases                  V(s)=0 V(b)=0 V(d)=1.0
338  biasList                V(g)=0.6,0.8,1.0
339  freq                    dec 10 1e3 1e8
340  outputs                 N(d)
341  instanceParameters      w=1.2e-6 l=1e-6 mult_fn=2
342  modelParameters         parameters/nmosParameters
343  modelParameters         rbodymod=1 rgatemod=2 rdsmod=1
```

MULT_FN

Similarly, 7 corresponding QA tests were added for pmos case as well

Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

11) 2023bug1: BSIM4 non-zero terminal current with all terminal voltages zero (NXP/Cadence)

Old Code

b4ld.c

```
T2 = Vgs_eff * Vaux;
dT2_dVg = dVgs_eff_dVg * Vaux + Vgs_eff * dVaux_dVg;
dT2_dVd = Vgs_eff * dVaux_dVd;
dT2_dVb = Vgs_eff * dVaux_dVb;


T2 = (Vgs_eff - Vbseff) * Vaux;
dT2_dVg = dVgs_eff_dVg * Vaux + (Vgs_eff - Vbseff) * dVaux_dVg;
dT2_dVb = -Vaux + (Vgs_eff - Vbseff) * dVaux_dVb;


T2 = (Vgs_eff - Vbseff) * Vaux;
dT2_dVg = dVgs_eff_dVg * Vaux + (Vgs_eff - Vbseff) * dVaux_dVg;
dT2_dVd = (Vgs_eff - Vbseff) * dVaux_dVd;
dT2_dVb = -Vaux + (Vgs_eff - Vbseff) * dVaux_dVb;
```

New Code

b4ld.c

```
T2 = Vgs * Vaux;
dT2_dVg = Vaux + Vgs * dVaux_dVg;
dT2_dVd = Vgs * dVaux_dVd;
dT2_dVb = Vgs * dVaux_dVb;


T2 = (Vgs - Vbs) * Vaux;
dT2_dVg = Vaux + (Vgs - Vbs) * dVaux_dVg;
dT2_dVb = -Vaux + (Vgs - Vbs) * dVaux_dVb;

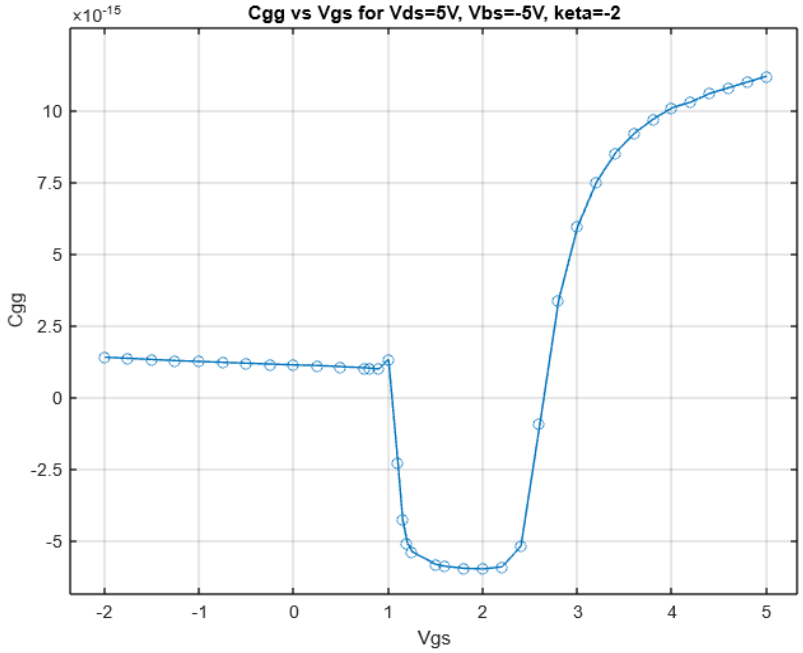

T2 = (Vgs - Vbs) * Vaux;
dT2_dVg = Vaux + (Vgs - Vbs) * dVaux_dVg;
dT2_dVd = (Vgs - Vbs) * dVaux_dVd;
dT2_dVb = -Vaux + (Vgs - Vbs) * dVaux_dVb;
```

12) 2023bug2: Negative Cgg/Cbg at high Vdd, high back bias and negative keta (TI/Cadence)

Issue: Negative cgg/cbg is encountered in BSIM4 4.8.2 when model is driven to high vdd and high back bias with negative keta:

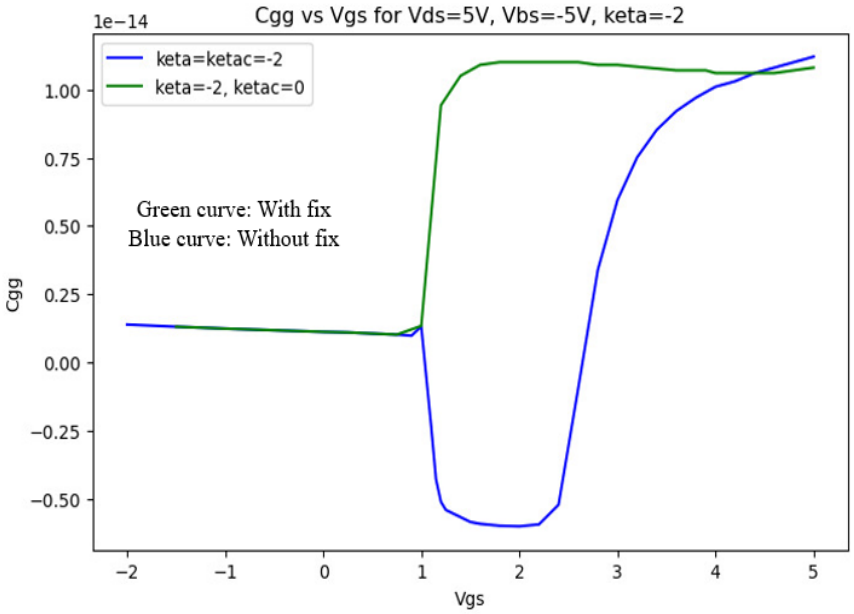
Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/



BSIM4 4.8.2

Solution: A new parameter ketac is introduced in BSIM4 4.8.3 with ketac = -0.047V^{-1} as default value to decouple the dc/cv behavior:



BSIM4 4.8.3

Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

New Code

bsim4def.h

```
double BSIM4ketac;

double BSIM4ketac;

double BSIM4lketac;

double BSIM4wketac;

double BSIM4pketac;

unsigned  BSIM4ketacGiven   :1;

unsigned  BSIM4lketacGiven   :1;

unsigned  BSIM4wketacGiven   :1;

unsigned  BSIM4pketacGiven   :1;


#define BSIM4_MOD_KETAC                 1276
#define BSIM4_MOD_LKETAC                1277
#define BSIM4_MOD_WKETAC                1278
#define BSIM4_MOD_PKETAC                1279
```

New Code

b4.c

```
IOP( "ketac", BSIM4_MOD_KETAC, IF_REAL, "Body-bias coefficient of non-uniform depletion width effect in dynamic evaluation."),

IOP( "lketac", BSIM4_MOD_LKETAC, IF_REAL, "Length dependence of ketac"),

IOP( "wketac", BSIM4_MOD_WKETAC, IF_REAL, "Width dependence of ketac"),

IOP( "pketac", BSIM4_MOD_PKETAC, IF_REAL, "Cross-term dependence of ketac"),
```

New Code

b4set.c

```
if (!model->BSIM4ketacGiven)
    model->BSIM4ketac = -0.047;     /* unit  / V */

if (!model->BSIM4lketacGiven)
    model->BSIM4lketac = 0.0;

if (!model->BSIM4wketacGiven)
    model->BSIM4wketac = 0.0;

if (!model->BSIM4pketacGiven)
    model->BSIM4pketac = 0.0;
```

New Code

b4mask.c

```
case BSIM4_MOD_KETAC:
    value->rValue = model->BSIM4ketac;
    return(OK);

case BSIM4_MOD_LKETAC:
    value->rValue = model->BSIM4lketac;
    return(OK);
```

Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

```
case BSIM4_MOD_WKETAC:
    value->rValue = model->BSIM4wketac;
    return(OK);

case BSIM4_MOD_PKETAC:
    value->rValue = model->BSIM4pketac;
    return(OK);
```

New Code

b4mpar.c

```
case BSIM4_MOD_KETAC:
    mod->BSIM4ketac = value->rValue;
    mod->BSIM4ketacGiven = TRUE;
    break;

 case BSIM4_MOD_LKETAC:
    mod->BSIM4lketac = value->rValue;
    mod->BSIM4lketacGiven = TRUE;
    break;

 case BSIM4_MOD_WKETAC:
    mod->BSIM4wketac = value->rValue;
    mod->BSIM4wketacGiven = TRUE;
    break;

 case BSIM4_MOD_PKETAC:
    mod->BSIM4pketac = value->rValue;
    mod->BSIM4pketacGiven = TRUE;
    break;
```

New Code

b4temp.c

```
pParam->BSIM4ketac = model->BSIM4ketac
                   + model->BSIM4lketac * Inv_L
                   + model->BSIM4wketac * Inv_W
                   + model->BSIM4pketac * Inv_LW;
```

New Code

b4ld.c

```
double tmp, Abulk, Abulk_Q, dAbulk_dVb, dAbulk_Q_dVb, Abulk0, Abulk0_Q, dAbulk0_dVb, dAbulk0_Q_dVb;

double diffVds, diffVdsCV, dAbulk_dVg, dAbulk_Q_dVg;
```

Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

```
/* Calculate Abulk_Q */
    T9 = 0.5 * pParam->BSIM4k1ox * Lpe_Vb / sqrtPhis;
    T1 = T9 + here->BSIM4k2ox - pParam->BSIM4k3b * Vth_NarrowW;
    dT1_dVb = -T9 / sqrtPhis * dsqrtPhis_dVb;
    T9 = sqrt(pParam->BSIM4xj * Xdep);
    tmp1 = Leff + 2.0 * T9;
    T5 = Leff / tmp1;
    tmp2 = pParam->BSIM4a0 * T5;
    tmp3 = pParam->BSIM4weff + pParam->BSIM4b1;
    tmp4 = pParam->BSIM4b0 / tmp3;
    T2 = tmp2 + tmp4;
    dT2_dVb = -T9 / tmp1 / Xdep * dXdep_dVb;
    T6 = T5 * T5;
    T7 = T5 * T6;
    Abulk0_Q = 1.0 + T1 * T2;
    dAbulk0_Q_dVb = T1 * tmp2 * dT2_dVb + T2 * dT1_dVb;
    T8 = pParam->BSIM4ags * pParam->BSIM4a0 * T7;
    dAbulk_Q_dVg = -T1 * T8;
    Abulk_Q = Abulk0_Q + dAbulk_Q_dVg * Vgsteff;
    dAbulk_Q_dVb = dAbulk0_Q_dVb - T8 * Vgsteff * (dT1_dVb
                 + 3.0 * T1 * dT2_dVb);
    if (Abulk0_Q < 0.1) /* added to avoid the problems caused by Abulk0_Q */
    {   T9 = 1.0 / (3.0 - 20.0 * Abulk0_Q);
        Abulk0_Q = (0.2 - Abulk0_Q) * T9;
        dAbulk0_Q_dVb *= T9 * T9;
    }
    if (Abulk_Q < 0.1)
    {   T9 = 1.0 / (3.0 - 20.0 * Abulk_Q);
        Abulk_Q = (0.2 - Abulk_Q) * T9;
        T10 = T9 * T9;
        dAbulk_Q_dVb *= T10;
        dAbulk_Q_dVg *= T10;
    }
    here->BSIM4Abulk = Abulk_Q;
    T2 = pParam->BSIM4ketac * Vbseff;
    if (T2 >= -0.9)
    {   T0 = 1.0 / (1.0 + T2);
        dT0_dVb = -pParam->BSIM4ketac * T0 * T0;
    }
    else
    {   T1 = 1.0 / (0.8 + T2);
        T0 = (17.0 + 20.0 * T2) * T1;
        dT0_dVb = -pParam->BSIM4ketac * T1 * T1;
    }
    dAbulk_Q_dVg *= T0;
    dAbulk_Q_dVb = dAbulk_Q_dVb * T0 + Abulk_Q * dT0_dVb;
    dAbulk0_Q_dVb = dAbulk0_Q_dVb * T0 + Abulk0_Q * dT0_dVb;
    Abulk_Q *= T0;
    Abulk0_Q *= T0;
```

Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

13. 2024bug1: Making bug-fix (2023bug2) backward compatible (ADI/Cadence)

Issue: ketac was used to compute Abulk0_Q, whereas keta was used to compute Abulk0. Consequently, the results for version=4.8.2 (or earlier) that specifies keta, and version=4.8.3_beta1 that specifies ketac for the same model card, will NOT match, because Abulk0_Q will differ from Abulk0. This makes the change backward incompatible.

Solution:

- In BSIM4 4.8.3, the default of the ketac, lketac, wketac, pketac is set to keta, lketa, wketa, pketa correspondingly.
- In BSIM4 4.8.3_beta1, Abulk_Q calculation (which uses ketac) was an exact copy of the Abulk calculation (which uses keta) and Abulk_Q was never used. In the BSIM4 4.8.3, the value of Abulk0 is now captured before it gets multiplied by T0, and then T2 is computed.
- BSIM4 4.8.3 is now backward compatible with BSIM4 4.8.2.

Old Code

b4set.c

```
if (!model->BSIM4ketacGiven)
    model->BSIM4ketac = -0.047;    /* unit  / V */

if (!model->BSIM4lketacGiven)
    model->BSIM4lketac = 0.0;

if (!model->BSIM4wketacGiven)
    model->BSIM4wketac = 0.0;

if (!model->BSIM4pketacGiven)
    model->BSIM4pketac = 0.0;
```

b4ld.c

```
dAbulk_dVg *= T0;
dAbulk_dVb = dAbulk_dVb * T0 + Abulk * dT0_dVb;
dAbulk0_dVb = dAbulk0_dVb * T0 + Abulk0 * dT0_dVb;
Abulk *= T0;
Abulk0 *= T0;
```

Summary of changes for BSIM4 4.8.3 release

BSIM Group, UC Berkeley, IIT Kanpur
List of developers: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/

```
/* Calculate Abulk_Q */
    T9 = 0.5 * pParam->BSIM4k1ox * Lpe_Vb / sqrtPhis;
    T1 = T9 + here->BSIM4k2ox - pParam->BSIM4k3b * Vth_NarrowW;
    dT1_dVb = -T9 / sqrtPhis * dsqrtPhis_dVb;
    T9 = sqrt(pParam->BSIM4xj * Xdep);
    tmp1 = Leff + 2.0 * T9;
    T5 = Leff / tmp1;
    tmp2 = pParam->BSIM4a0 * T5;
    tmp3 = pParam->BSIM4weff + pParam->BSIM4b1;
    tmp4 = pParam->BSIM4b0 / tmp3;
    T2 = tmp2 + tmp4;
    dT2_dVb = -T9 / tmp1 / Xdep * dXdep_dVb;
    T6 = T5 * T5;
    T7 = T5 * T6;
    Abulk0_Q = 1.0 + T1 * T2;
    dAbulk0_Q_dVb = T1 * tmp2 * dT2_dVb + T2 * dT1_dVb;
    T8 = pParam->BSIM4ags * pParam->BSIM4a0 * T7;
    dAbulk_Q_dVg = -T1 * T8;
    Abulk_Q = Abulk0_Q + dAbulk_Q_dVg * Vgsteff;
    dAbulk_Q_dVb = dAbulk0_Q_dVb - T8 * Vgsteff * (dT1_dVb
                 + 3.0 * T1 * dT2_dVb);
    if (Abulk0_Q < 0.1) /* added to avoid the problems caused by Abulk0_Q */
    {   T9 = 1.0 / (3.0 - 20.0 * Abulk0_Q);
        Abulk0_Q = (0.2 - Abulk0_Q) * T9;
        dAbulk0_Q_dVb *= T9 * T9;
    }
    if (Abulk_Q < 0.1)
    {   T9 = 1.0 / (3.0 - 20.0 * Abulk_Q);
        Abulk_Q = (0.2 - Abulk_Q) * T9;
        T10 = T9 * T9;
        dAbulk_Q_dVb *= T10;
        dAbulk_Q_dVg *= T10;
    }
here->BSIM4Abulk = Abulk_Q;
T2 = pParam->BSIM4ketac * Vbseff;
if (T2 >= -0.9)
{   T0 = 1.0 / (1.0 + T2);
    dT0_dVb = -pParam->BSIM4ketac * T0 * T0;
}
else
{   T1 = 1.0 / (0.8 + T2);
    T0 = (17.0 + 20.0 * T2) * T1;
    dT0_dVb = -pParam->BSIM4ketac * T1 * T1;
}
dAbulk_Q_dVg *= T0;
dAbulk_Q_dVb = dAbulk_Q_dVb * T0 + Abulk_Q * dT0_dVb;
dAbulk0_Q_dVb = dAbulk0_Q_dVb * T0 + Abulk0_Q * dT0_dVb;
Abulk_Q *= T0;
Abulk0_Q *= T0;
```

New Code

b4set.c

```
if (!model->BSIM4ketacGiven)
    model->BSIM4ketac = model->BSIM4keta;     /* unit  / V */


if (!model->BSIM4lketacGiven)
    model->BSIM4lketac = model->BSIM4lketa;
```

```
if (!model->BSIM4wketacGiven)
    model->BSIM4wketac = model->BSIM4wketa;

if (!model->BSIM4pketacGiven)
    model->BSIM4pketac = model->BSIM4pketa;
```

b4ld.c

```
dAbulk_dVg *= T0;
dAbulk_dVb = dAbulk_dVb * T0 + Abulk * dT0_dVb;
dAbulk0_Q_dVb = dAbulk0_dVb; // copy before scaling
dAbulk0_dVb = dAbulk0_dVb * T0 + Abulk0 * dT0_dVb;
Abulk *= T0;
Abulk0_Q = Abulk0; // copy before scaling
Abulk0 *= T0;

/* Calculate Abulk0_Q */
if (pParam->BSIM4ketac != pParam->BSIM4keta) {
    T2 = pParam->BSIM4ketac * Vbseff;
    if (T2 >= -0.9)
    {   T0 = 1.0 / (1.0 + T2);
        dT0_dVb = -pParam->BSIM4ketac * T0 * T0;
    }
    else
    {   T1 = 1.0 / (0.8 + T2);
        T0 = (17.0 + 20.0 * T2) * T1;
        dT0_dVb = -pParam->BSIM4ketac * T1 * T1;
    }
    dAbulk0_Q_dVb = dAbulk0_Q_dVb * T0 + Abulk0_Q * dT0_dVb;
    Abulk0_Q *= T0;
} else {
    dAbulk0_Q_dVb = dAbulk0_dVb;
    Abulk0_Q = Abulk0;
}
```

14. 2024bug2: Interpretation for default value of dW & dL parameters in BSIM4_4.8.3_beta1 Technical Manual (GF)

Issue: In **BSIM4 4.8.3_beta1** Technical Manual, dW and dL parameters are defined in Section 1.3, Equation 1.12 as:

$$dW = dW' + DWG \cdot V_{gsteff} + DWB\left(\sqrt{\Phi_s - V_{bseff}} - \sqrt{\Phi_s}\right)$$

$$dW' = WINT + \frac{WL}{L^{WLN}} + \frac{WW}{W^{WWN}} + \frac{WWL}{L^{WLN} W^{WWN}}$$

$$dL = LINT + \frac{LL}{L^{LLN}} + \frac{LW}{W^{LWN}} + \frac{LWL}{L^{LLN} W^{LWN}}$$

And the default values are defined in A.16 as shown highlighted:

BSIM4 4.8.3_beta1 Technical Manual

Parameter name	Description	Default name	Binnable?	Note
WL	Coefficient of length dependence for width offset	0.0mWLN	No	-
WLN	Power of length dependence of width offset	1.0	No	-
WW	Coefficient of width dependence for width offset	0.0mWWN	No	-
WWN	Power of width dependence of width offset	1.0	No	-
WWL	Coefficient of length and width cross term dependence for width offset	0.0 mWWN+WLN	No	-
LL	Coefficient of length dependence for length offset	0.0mLLN	No	-
LLN	Power of length dependence for length offset	1.0	No	-
LW	Coefficient of width dependence for length offset	0.0mLWN	No	-
LWN	Power of width dependence for length offset	1.0	No	-
LWL	Coefficient of length and width cross term dependence for length offset	0.0 mLWN+LLN	No	-

But the code simply defines these default values as **0.0 without any dependence on WLN or WWN.** And similarly for other highlighted parameters.

Solution: **BSIM4 4.8.3** Technical Manual is updated by setting the default values of all the highlighted parameters = 0.0. Also, the term "Default name" is changed to "Default value":

BSIM4 4.8.3 Technical Manual

Parameter name	Description	Default value	Binnable?	Note
WL	Coefficient of length dependence for width offset	0.0	No	-
WLN	Power of length dependence of width offset	1.0	No	-
WW	Coefficient of width dependence for width offset	0.0	No	-
WWN	Power of width dependence of width offset	1.0	No	-
WWL	Coefficient of length and width cross term dependence for width offset	0.0	No	-
LL	Coefficient of length dependence for length offset	0.0	No	-
LLN	Power of length dependence for length offset	1.0	No	-
LW	Coefficient of width dependence for length offset	0.0	No	-
LWN	Power of width dependence for length offset	1.0	No	-
LWL	Coefficient of length and width cross term dependence for length offset	0.0	No	-

15. 2024enh1: Missing Details of Length/Width/Cross dependencies of binnable model parameters in BSIM4_4.8.3_beta1 Technical Manual (GF)

Issue: The detail for L/W/P dependencies (e.g. vth0-lvth0/wvth0/pvth0) of binnable model parameters is missing from the technical manual.

Solution: An extra note in A.18 of **BSIM4_4.8.3 Technical Manual** is added as follows:

Note-11: Each binnable model parameter M_i is calculated as a function of M, a length dependent term, LM, a width dependent term, WM and a product LW term, PM:

$$M_i = M + \frac{LM}{L_{eff}} + \frac{WM}{W_{eff}} + \frac{PM}{L_{eff}W_{eff}}$$

Default values of all l*/w*/p* parameters are zero.

And consequently Note-11 is added to the "Note" column of all the binnable parameters.

16. 2024bug3: Missing lines in b4mask.c and b4ask.c to handle a few parameters (ADI)

Issue: The b4ask.c is missing lines for QINV and b4mask.c file is missing lines for 11 parameters (NGCON, LINTNOI, LLAMBDA, LVTL, LXN, WLAMBDA, WVTL, WXN, PLAMBDA, PVTL, PXN).

Solution: The missing lines in the two files (b4ask.c and b4mask.c) are added to handle the twelve parameters mentioned below.

New Code

b4mask.c

```
case BSIM4_MOD_NGCON:
    value->rValue = model->BSIM4ngcon;
    return(OK);

case BSIM4_MOD_LINTNOI:
    value->rValue = model->BSIM4lintnoi;
    return(OK);

case BSIM4_MOD_LLAMBDA:
    value->rValue = model->BSIM4llambda;
    return(OK);
case BSIM4_MOD_LVTL:
    value->rValue = model->BSIM4lvtl;
    return(OK);
case BSIM4_MOD_LXN:
    value->rValue = model->BSIM4lxn;
    return(OK);
```

```
case BSIM4_MOD_WLAMBDA:
    value->rValue = model->BSIM4wlambda;
    return(OK);
case BSIM4_MOD_WVTL:
    value->rValue = model->BSIM4wvtl;
    return(OK);
case BSIM4_MOD_WXN:
    value->rValue = model->BSIM4wxn;
    return(OK);

case BSIM4_MOD_PLAMBDA:
    value->rValue = model->BSIM4plambda;
    return(OK);
case BSIM4_MOD_PVTL:
    value->rValue = model->BSIM4pvtl;
    return(OK);
case BSIM4_MOD_PXN:
    value->rValue = model->BSIM4pxn;
    return(OK);
```

New Code

b4ask.c

```
case BSIM4_QINV:
    value->rValue = here->BSIM4qinv;
    return(OK);
```

17. 2024bug4: BSIM4 version check (ADI)

In b4set.c the version number needs to be updated to 4.83.

Old Code

b4set.c

```
if (!model->BSIM4versionGiven)
   model->BSIM4version = 4.82;
```

New Code

b4set.c

```
if (!model->BSIM4versionGiven)
   model->BSIM4version = 4.83;
```

In b4check.c the version number needs to be updated to BSIM4.8.3.

Old Code	**b4check.c**
	`if (fabs(model->BSIM4version - 4.82) > 0.0001){`
	`  fprintf(fplog, "Warning: This model is BSIM4.8.2; you specified a wrong version number.\n");`
	`  printf("Warning: This model is BSIM4.8.2; you specified a wrong version number.\n");`

New Code	**b4check.c**
	`if (fabs(model->BSIM4version - 4.83) > 0.0001){`
	`  fprintf(fplog, "Warning: This model is BSIM4.8.3; you specified a wrong version number.\n");`
	`  printf("Warning: This model is BSIM4.8.3; you specified a wrong version number.\n");`

18. 2024bug5: Code update for developer's names (ADI)

We have updated the following section of the code in b4check.c file to make the developers list updated

Old Code	**b4check.c**
	`fprintf(fplog, "BSIM4: Berkeley Short Channel IGFET Model-4\n");`
	`fprintf(fplog, "Developed by Xuemei (Jane) Xi, Mohan Dunga, Prof. Ali Niknejad and Prof. Chenming Hu in 2003.\n");`
	`fprintf(fplog, "\n");`

New Code	**b4check.c**
	`fprintf(fplog, "BSIM4: Berkeley Short Channel IGFET Model-4\n");`
	`fprintf(fplog, "Developers list: https://www.bsim.berkeley.edu/models/bsim4/auth_bsim4/ \n");`
	`fprintf(fplog, "\n");`